

device-specific identity

pattern energy

device-specific identity

Real image

Latent autoencoder

LoRA objective

**Text-to-image
generation**

$R_{\text{real}} = 3.57 \times 10^{-2}$
same-model AUC 1.000

$R_{\text{VAE}} = 1.31 \times 10^{-2}$
 $\eta = 0.366$, AUC 0.981

measurable from $\alpha \approx 3$
scalar-loss response not
device- or alignment-specific

$\lambda_U \leq 0.15\%$
 $\tau_U \leq 0.41\%$